3.5 D: Sequences and series

	Content
D1	Understand and use the binomial expansion of $(a + bx)^n$ for positive integer n; the notations $n!$, $n\mathrm{C}r$ and $\begin{pmatrix} n \\ r \end{pmatrix}$; link to binomial probabilities. Extend to any rational n, including its use for approximation; be aware that the expansion is valid for $\left\lvert \frac{bx}{a} \right\rvert < 1$. (Proof not required.)

	Content
D2	Work with sequences including those given by a formula for the n th term and those generated by a simple relation of the form $x_{n+1} = \mathrm{f}(x_n)$; increasing sequences; decreasing sequences; periodic sequences.

	Content
D3	Understand and use sigma notation for sums of series.

	Content
D4	Understand and work with arithmetic sequences and series, including the formulae for n th term and the sum to n terms.

	Content
D5	Understand and work with geometric sequences and series including the formulae for the n th term and the sum of a finite geometric series; the sum to infinity of a convergent geometric series, including the use of $\lvert r \rvert < 1$; modulus notation.

	Content
D6	Use sequences and series in modelling.